UE24CS352A · Machine Learning Mini-Project

Video Analysis of a Fruit-Fly Pair

Position, sex, facing direction and male wing angles

Reimplementation of Steven Herbst's CS229 pipeline (2018) · [Your name] · [Teammate name]

Task and dataset

- One male and one female fruit fly in a circular well.
- Frames hand-labeled with heads, abdomens, body points and male wing tips.
- Outputs: annotated video, detection CSV and wing-angle plot.

326
labeled frames from one recording

218 / 108
train / test split, seed 0

5
video clips, 1530 × 1530 at 35 fps

Dataset and annotations: Herbst's CS229 project (Stanford, Clandinin lab). Pipeline reimplemented in Python.

Four-stage method

STAGE 1

Body detection

Threshold the frame. A decision tree on contour area says zero, one or two flies. One-fly contours give positions.

STAGE 2

Sex

Logistic regression on both contours' area and aspect ratio. Runs only when two separate flies remain.

STAGE 3

Heading

Image moments give the body axis. HOG, 15 PCA components and logistic regression resolve the 180° flip.

STAGE 4

Wings

Rotate the male upright. HOG, 40 PCA components and linear regression per wing half, clipped to 0–90°.

Individual model results

Stage	Test metric	Result	Examples
Body count	Classification error	0%	329 contours
Sex	Classification error	0%	97 pairs
Heading reversal	Classification error	0%	152 flies
Wing angle	Residual SD	7.26°	158 wings

Each stage is tested alone. The wing model is aligned with annotated headings, so these numbers do not include errors from earlier stages.

Complete pipeline results

194 of 216 labeled flies matched across 108 held-out test frames.

100%
sex accuracy on 194 matches

1.60°
mean heading error on 152 labeled headings

5.87°
mean absolute wing error on 158 wings

The 22 unmatched flies are excluded; a detection counts if it lies within 80 pixels of a body label. Herbst reports wing σ = 2.92° on his own split, so the two are not directly comparable.

Demo: annotated clip

What you see

- Outline per fly: orange-red for male, green for female
- Arrow: facing direction, from abdomen to head
- Label: M L## R## gives left and right wing angles

What it writes

- demo.mp4: the annotated video
- tracks.csv: frame, sex, x, y, heading, wing angles
- wing_trace.png: male wing angles over time

Play the clip (test4.mp4), then show the wing trace.

Runtime and limits

66 fps

on test4.mp4, against a 35 fps source (about 1.9× real time)

- Timing includes video I/O, annotation and CSV writing, not live-camera latency.
- Touching flies merge into one blob and are skipped. There is no tracking between frames.
- Train and test frames share one recording. New pairs and conditions are untested.
- Wing angles have no independently validated song labels.

Conclusion and future work

What we built

- A four-stage classical ML pipeline, reimplemented from Herbst (2018)
- 66 fps on 35 fps video; sex 100%, heading 1.6°, wings 5.9° error
- An annotated video, a CSV and a wing-angle plot

Next steps

- Handle touching flies instead of skipping them
- Add tracking between frames
- Test on new recordings; add leg tracking, as Herbst proposes

References: S. Herbst, Real-time Detailed Video Analysis of Fruit Flies, CS229 Final Project, Stanford, 2018 (github.com/sgherbst/cs229-project).
N. Dalal and B. Triggs, Histograms of Oriented Gradients for Human Detection, CVPR 2005.